

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 10%

QUESTIONS		
Q. No.	Question	Bloom's Level
1	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand
2	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand
3	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand
4	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember
5	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



SIMATS
ENGINEERING



SIMATS
Savitribha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)



TECHNICAL PRESENTATION

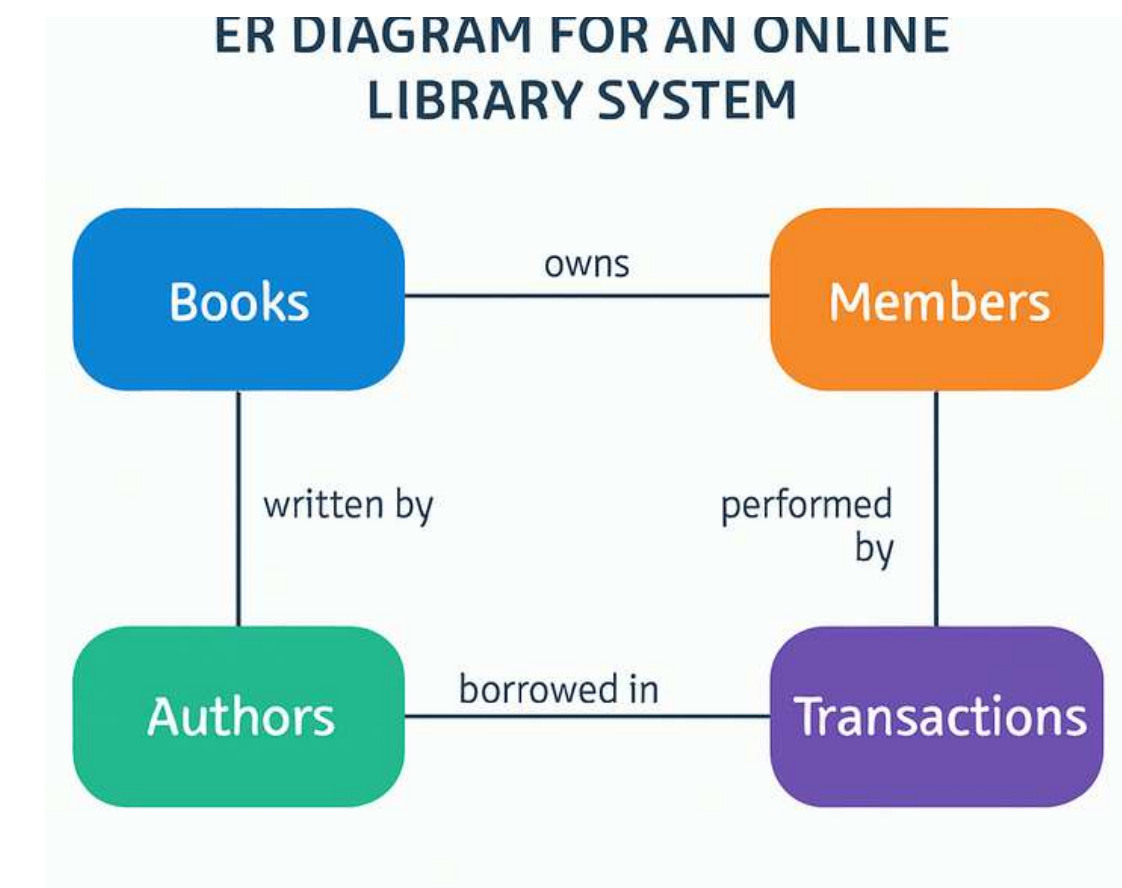
CSA0509-DATABASE MANAGEMENT SYSTEM

BY,
SANDRA SUDEVAN (192424422)

INTRODUCTION

1. ER Diagram for an Online Library System

- ER Diagram represents entities, attributes, and relationships.
- Used to design a database before implementation.
- Helps organize library data efficiently.





Problem Statement & Components



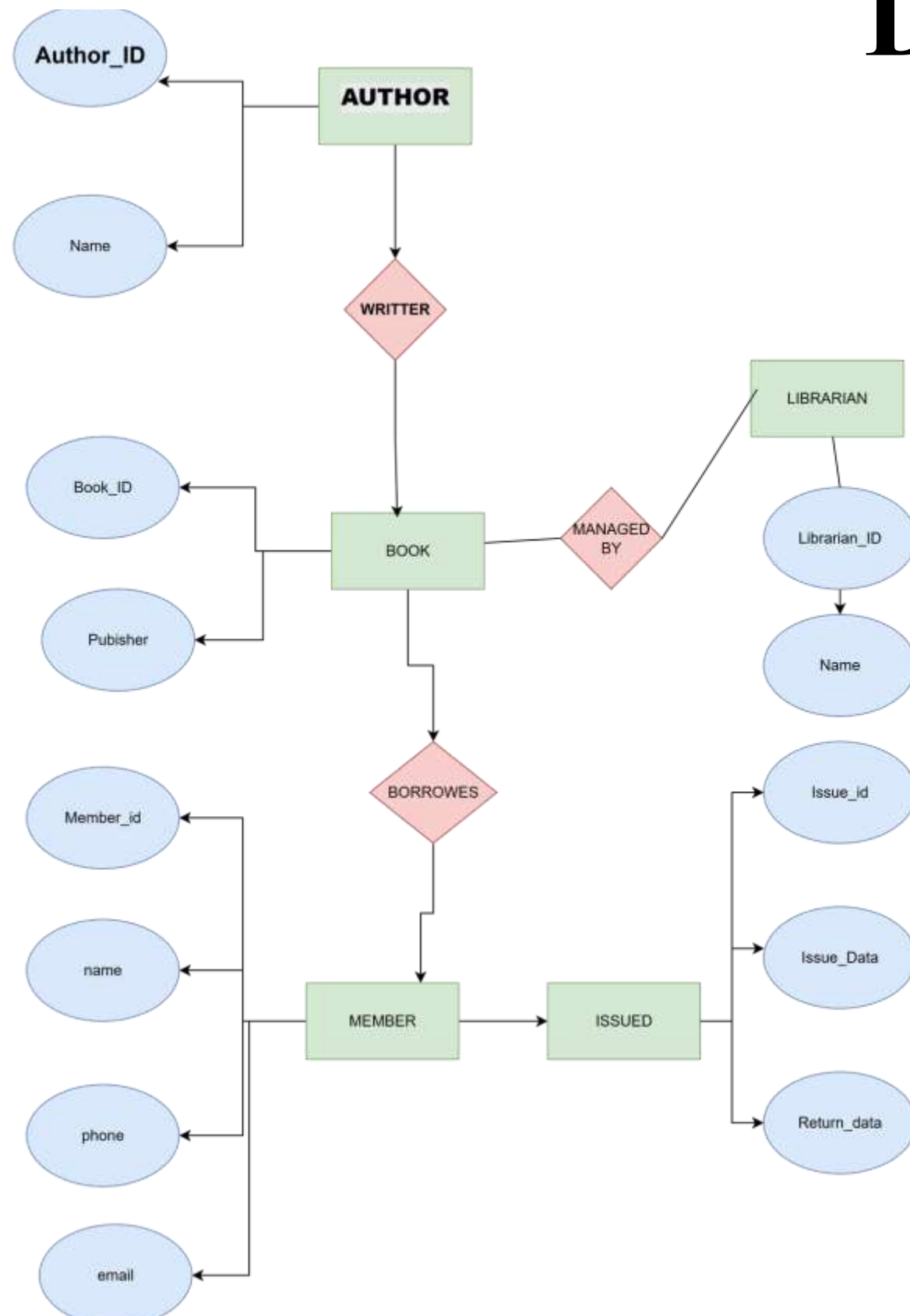
Problem Statement

- Manual library management is difficult.
- Tracking books and members takes time.
- Need an efficient database system.

Components

- Entities: Book, Member, Librarian, Author, Issue
- Attributes: Book_ID, Title, Member_ID, Issue_Date
- Relationships: Member borrows Book, Librarian manages Issue

ER Diagram



- Rectangles (Entities)
 - Diamonds (Relationships)
- Ovals (Attributes)



Advantages & Conclusion



Advantages

- Reduces data redundancy
- Improves data consistency
- Easy database maintenance
- Faster information retrieval

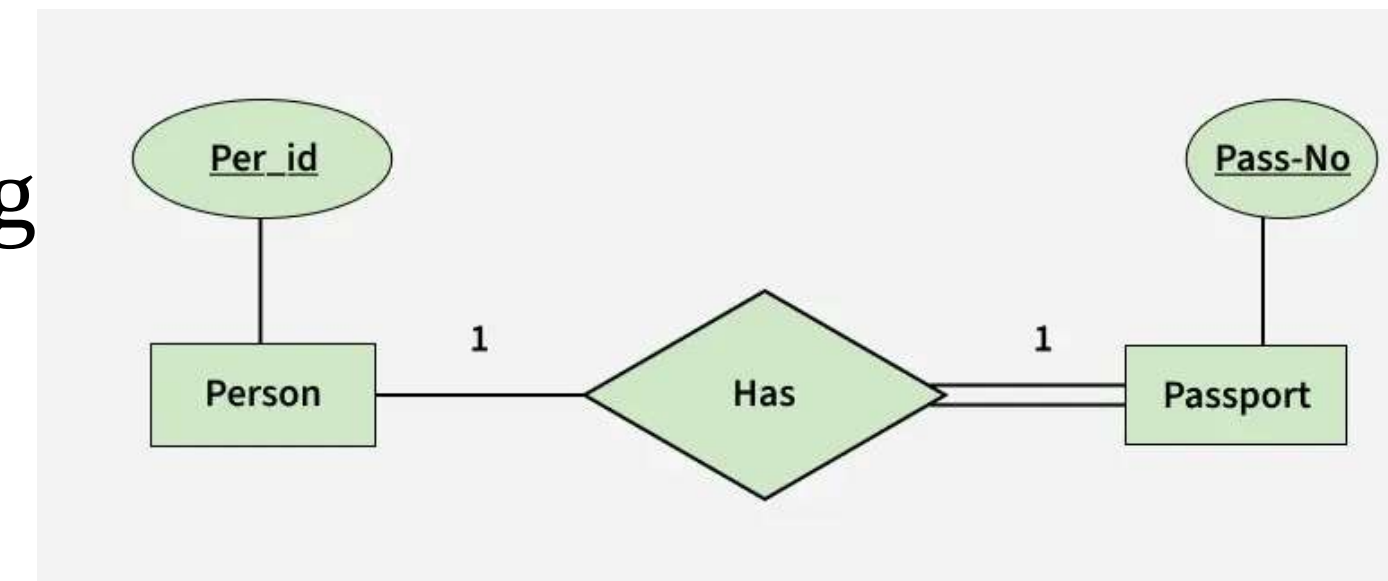
Conclusion

- ER diagrams provide a clear blueprint for developing an Online Library Management System.

EER Model – University Case Study

Enhanced Entity-Relationship (EER) Model

- EER is an advanced version of the ER model.
- It represents complex database structures using inheritance.
- It introduces generalization and specialization.
- Used in systems like universities, hospitals, and banks.





Generalization and Specialization

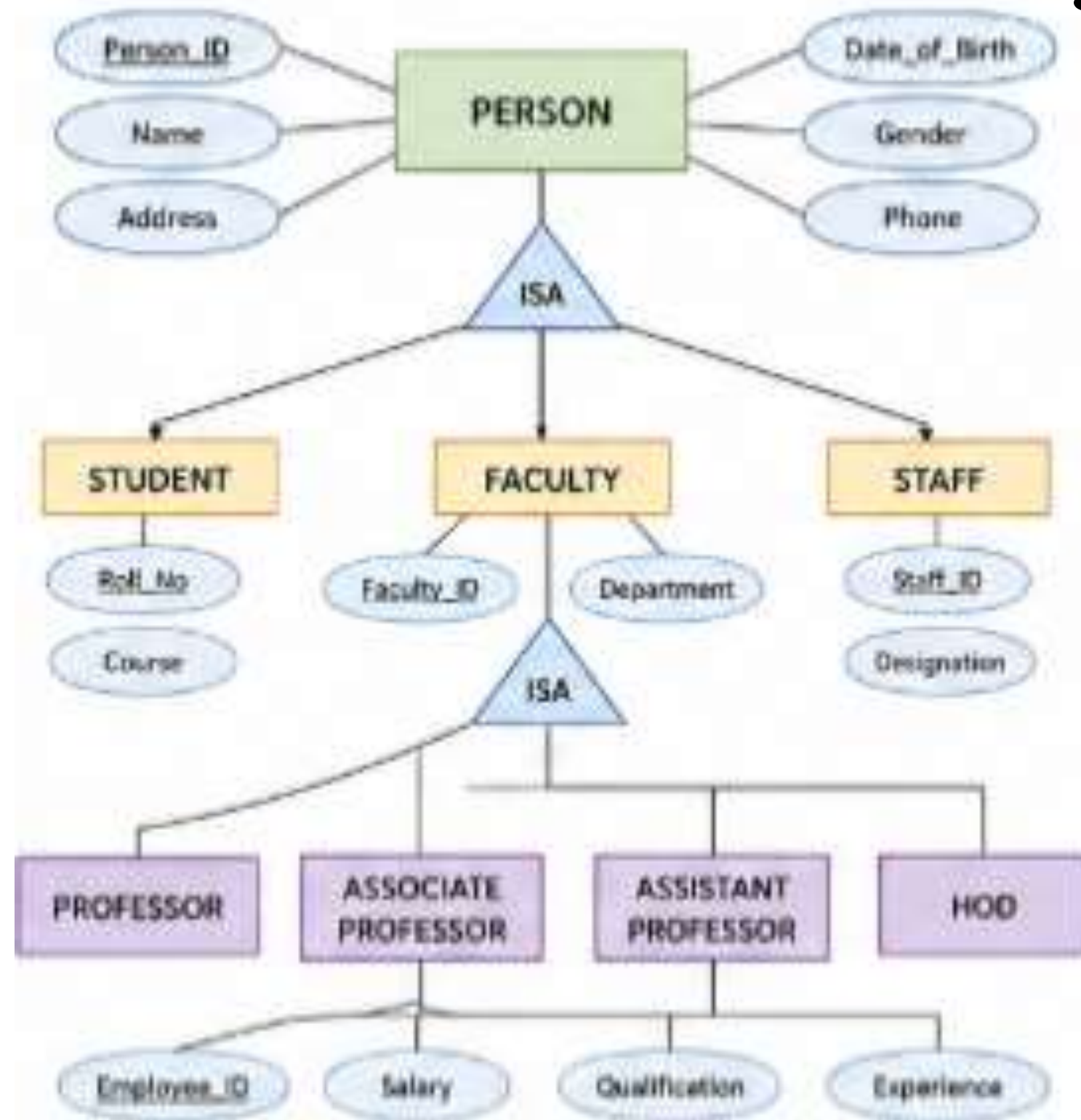
Generalization

- Combines similar entities into one superclass.
- Example: Student, Faculty, Staff → Person.
- Common attributes: Person_ID, Name, Address.

Specialization

- Divides a superclass into subclasses.
- Example: Faculty → Professor, Assistant Professor, HOD.
- Each subclass has its own specific attributes.

University EER Diagram



University Case Study

- Superclass: Person
- Subclasses: Student, Faculty, Staff
- Faculty Specialization: Professor, Assistant Professor, HOD

Benefits

- Reduces data redundancy.
- Supports inheritance.
- Organizes data efficiently



Applications and Conclusion



Applications

- University Management System
- Hospital Management
- Banking System
- Human Resource Management

Conclusion

- EER improves the ER model by supporting generalization and specialization.
- It provides a flexible and efficient database design for complex applications.

Data Independence in DBMS

- Data independence is the ability to modify the database schema without affecting application programs.
- It provides flexibility and simplifies database maintenance.
- It is achieved using the three-schema architecture in DBMS.





Types of Data Independence



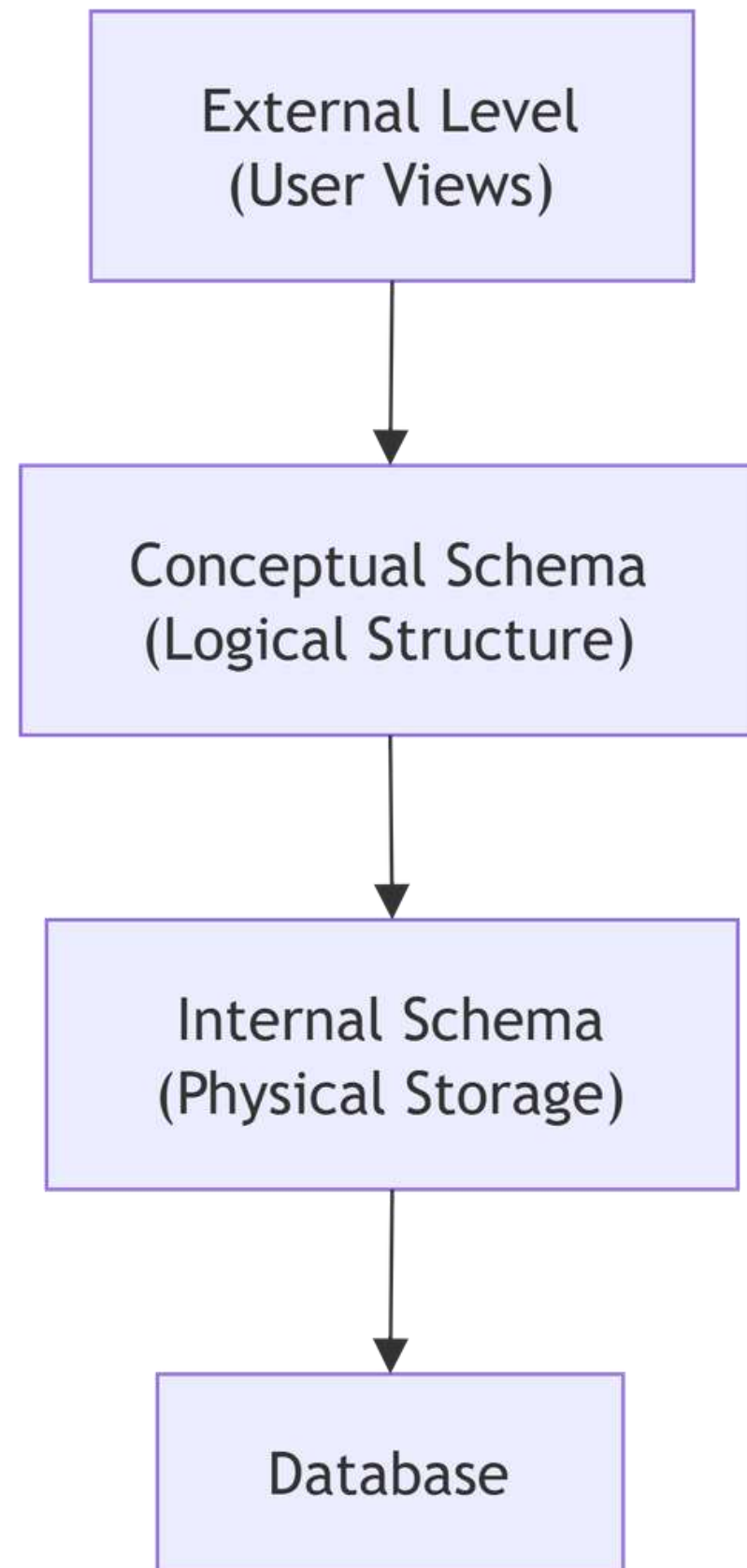
1. Logical Data Independence

- Changes can be made to the conceptual schema without affecting user applications.
- Example: Adding a new column to a table.

2. Physical Data Independence

- Changes to the internal storage do not affect the conceptual schema.
- Example: Changing the storage device or file organization.

Three-Schema Architecture



DBMS Architecture

- External Level – User views of the database.
- Conceptual Level – Overall logical structure of the database.
- Internal Level – Physical storage of data.

Advantages

- Data security
- Easy maintenance
- Better flexibility



Significance and Conclusion



Significance

- Reduces the impact of database changes.
- Improves scalability and flexibility.
- Simplifies application development.
- Enhances data management efficiency.

Conclusion

- Data independence is an important feature of DBMS that allows changes to the database structure without affecting applications, making systems more reliable and easier to maintain.

Software Components of DBMS

- A Database Management System (DBMS) consists of software components that manage data efficiently.
- These components process queries, store data, and ensure secure transactions.
- The three main components are Query Processor, Storage Manager, and Transaction Manager.





Main Components



1. Query Processor

- Interprets and executes SQL queries.
- Optimizes queries for better performance.

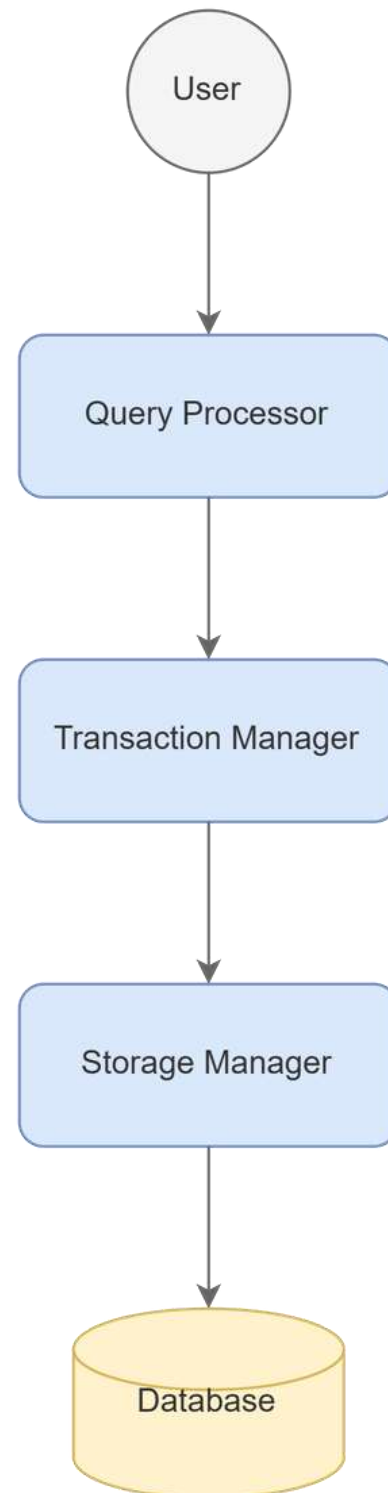
2. Storage Manager

- Manages data storage on disk.
- Controls files, indexes, and memory buffers.

3. Transaction Manager

- Ensures data consistency and reliability.
- Supports ACID properties and recovery.

DBMS Architecture



Functions

- Processes user requests.
- Stores and retrieves data.
- Maintains secure and reliable transactions.



Advantages and Conclusion



Advantages

- Fast query processing.
- Efficient data storage.
- Secure transaction management.
- Improves database performance.

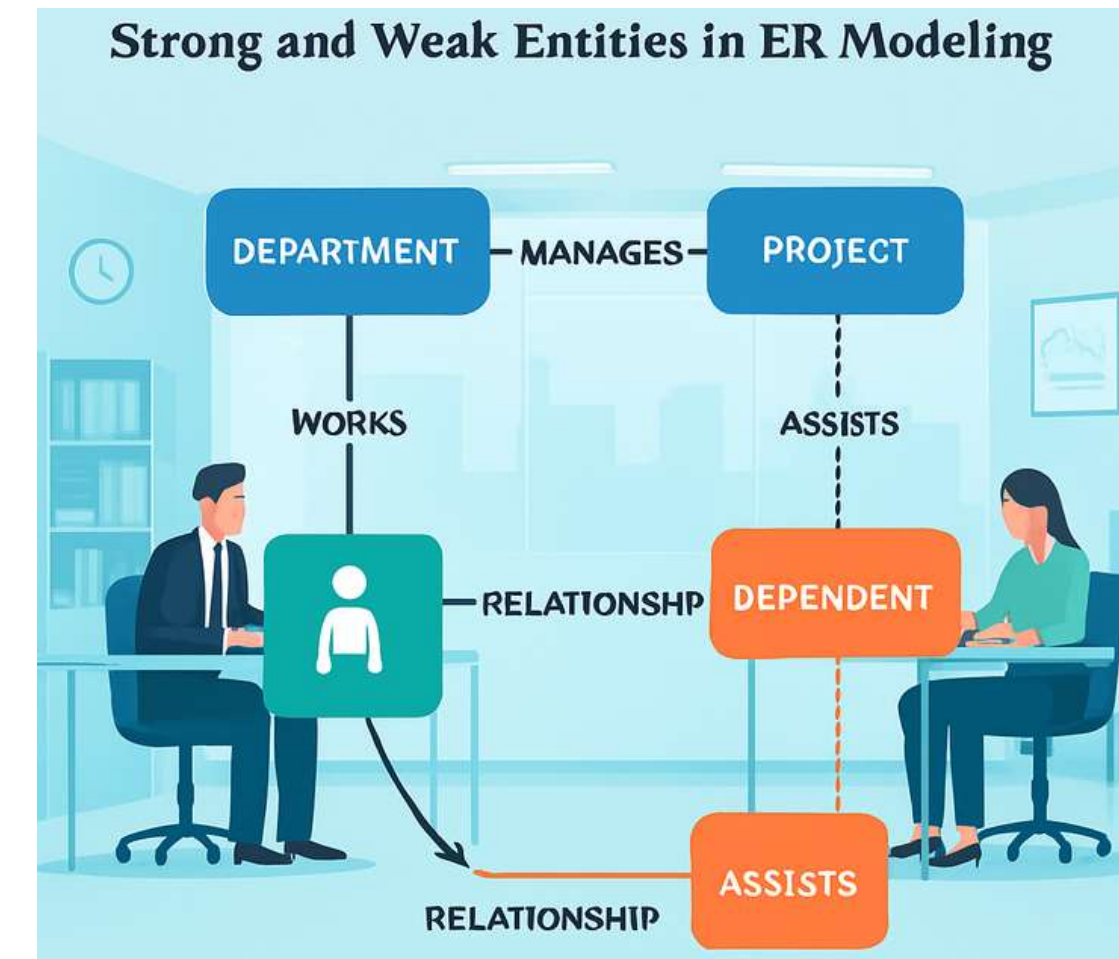
Conclusion

- The Query Processor, Storage Manager, and Transaction Manager work together to ensure efficient, secure, and reliable database operations in a DBMS.

Strong and Weak Entities in ER Modeling

Strong and Weak Entities in ER Modeling

- An entity represents a real-world object in a database.
- Entities are classified as Strong Entities and Weak Entities.
- They help define relationships and organize data efficiently in an ER model.





Strong Entity vs Weak Entity



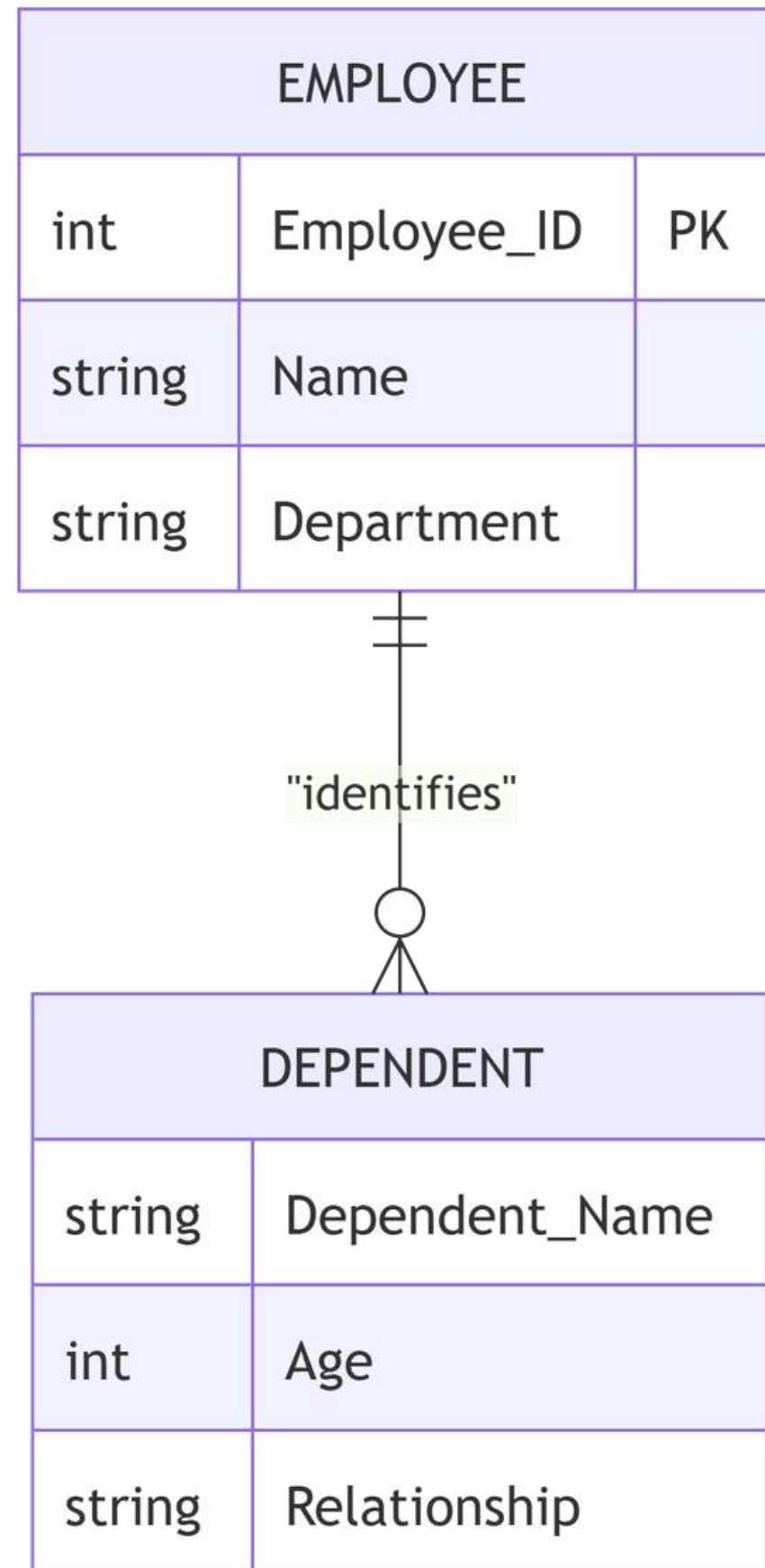
Strong Entity

- Has its own Primary Key.
- Exists independently.
- Represented by a single rectangle.
- Example: Student, Employee, Customer

Weak Entity

- Does not have a Primary Key.
- Depends on a strong entity for identification.
- Represented by a double rectangle.
- Example: Dependent, Order Item, Nominee

ER diagram showing Employee and Dependent



Strong Entity

- Has Primary Key
- Independent
- Single Rectangle
- Can exist alone

Weak Entity

- No Primary Key
- Depends on Strong Entity
- Double Rectangle
- Cannot exist alone



Applications and Conclusion



Applications

- Banking Systems
- Hospital Management
- Library Management
- Employee Database Systems

Conclusion

- Strong entities have their own identity and can exist independently.
- Weak entities rely on strong entities for identification.
- Understanding these concepts helps in designing accurate and efficient ER models.



THANK YOU